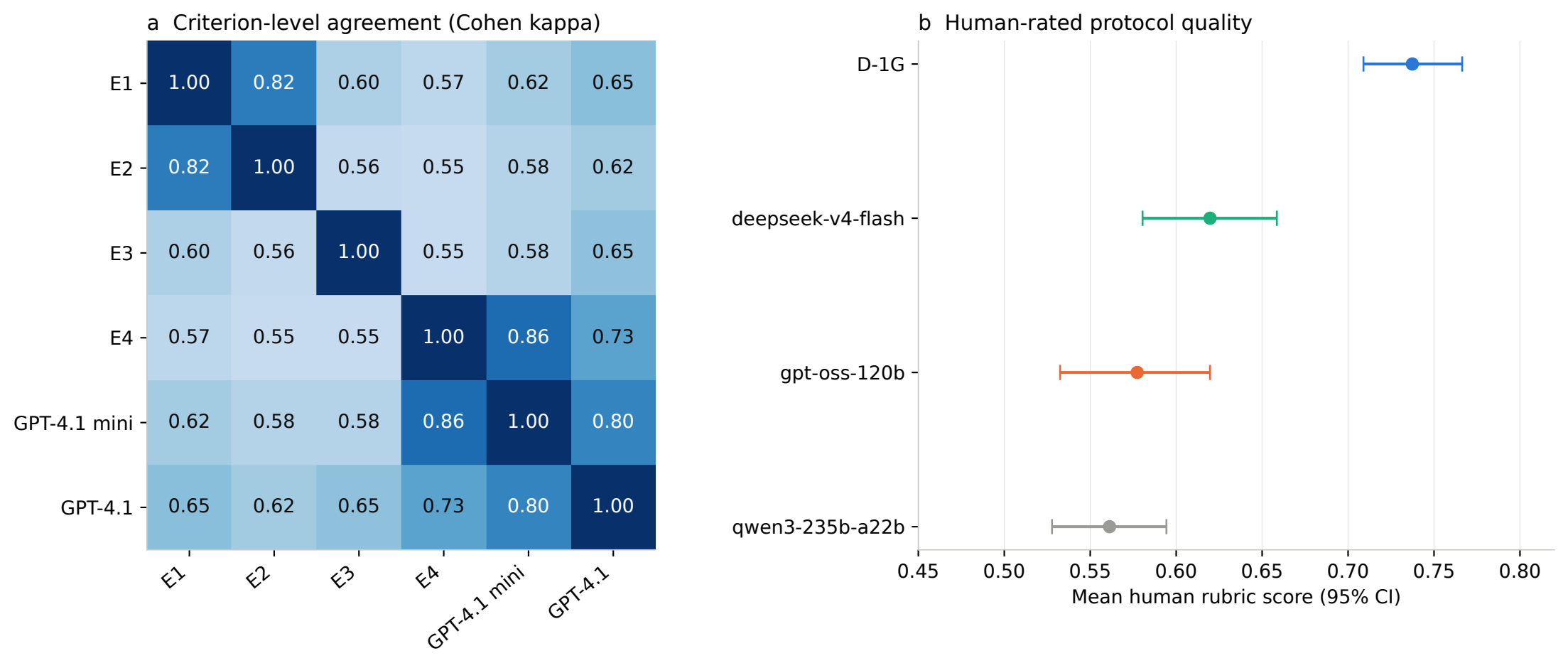
a Criterion-level agreement (Cohen kappa)
E1
E2
E3
E4
GPT-4.1 mini
GPT-4.1
1.00 0.82 0.60 0.57 0.62 0.65
0.82 1.00 0.56 0.55 0.58 0.62
0.60 0.56 1.00 0.55 0.58 0.65
0.57 0.55 0.55 1.00 0.86 0.73
0.62 0.58 0.58 0.86 1.00 0.80
0.65 0.62 0.65 0.73 0.80 1.00
b Human-rated protocol quality
D-1G
deepseek-v4-flash
gpt-oss-120b
qwen3-235b-a22b
0.45 0.50 0.55 0.60 0.65 0.70 0.75 0.80
Mean human rubric score (95% CI)